

FAR SIDE northern hemisphere

Pascal Crater



TYPE Impact crater
AGE About 4.1 billion years
DIAMETER 71 miles (115 km)

This is one of 300 lunar craters named after mathematicians. It honors the Frenchman Blaise Pascal. The image below was taken in 2004, with the camera looking directly down into the crater. The Sun is low in the sky, below the bottom of the picture. The tiny craters around Pascal are bowl-shaped and young, with circular rims much sharper than the older rim of Pascal. The larger crater's rim was initially eroded by slumping and rock slides and is now being worn down further by more recent impacts.



PASCAL AND ITS YOUNGER NEIGHBORS

FAR SIDE southern hemisphere

Tsiolkovsky Crater



TYPE Impact crater
AGE About 4.2 billion years
DIAMETER 123 miles (198 km)

Only half the size of Mare Crisium, this far-side crater is special because only half the interior basin has been filled with lava. The central peak is also unusually offset from the center of the crater. There have been extensive rock avalanches down the southern rim

ORBITER 3 IMAGE

The crest of the rim of Tsiolkovsky Crater runs to the upper right of this image. The diagonal banding to its right is probably the result of a large avalanche down the slope of the rim.



DARK FLOOR

If Tsiolkovsky had been formed earlier in lunar history, the volcanic activity would have been greater and more of the crater floor would have been filled with lava.

of the crater. The first images of the lunar far side was obtained in October 1959 by the Soviet spacecraft Luna 3. Resolution was low, but the features that could be seen were nevertheless given names, such as Mare Moscoviense and Sinus Astronautarum. Only a few craters could be made out, including this one.

Konstantin Tsiolkovsky was a Russian rocketry pioneer who not only designed a liquid hydrogen/liquid oxygen rocket but also suggested the multistage approach to spaceflight. The crater was penciled in as a possible landing site for one of the post-Apollo 17 missions, which were canceled.

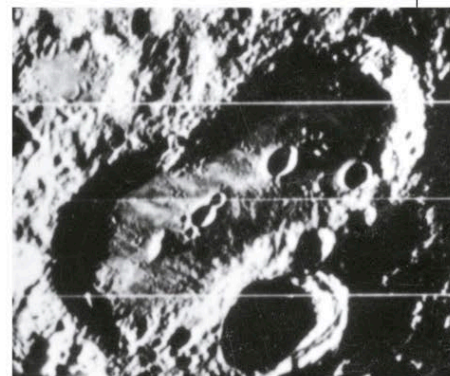
FAR SIDE southern hemisphere

Van de Graaff Crater



TYPE Double impact crater
AGE About 3.6 billion years
LENGTH 155 miles (250 km)

Less than 1 percent of the lunar craters are noncircular. Van de Graaff is typical of such irregular craters, which are produced on the rare occasions when the impacting asteroid hits the surface at an angle of less than 4°. Van de Graaff is also special because it is both magnetic and has the highest concentration of natural radiation. Most of the ancient lunar magnetic field decayed away over 3 billion years ago. However, there are still a few magnetic anomalies (magcons), of which Van de Graaff and nearby Aitken are the strongest. Magcons were discovered by small magnetometer subsatellites released by Apollos 15 and 16.



IRREGULARLY SHAPED CRATER

FAR SIDE southern hemisphere

Korolev Crater



TYPE Ringed impact crater
AGE About 3.7 billion years
DIAMETER 250 miles (405 km)

Sergei Korolev led the Soviet space effort in the 1950s and 1960s and was responsible for the early Sputnik and Vostok spacecraft. He has two craters named after him, one on the Moon and the other on Mars. Korolev is one of only 10 craters on the lunar far side that are more than 125 miles (200 km) across. It is double-ringed and pocked with smaller craters. The outer ring is 252 miles (405 km) in diameter. The inner ring is much less distinct. It is only half the height of the outer ring and its diameter is half that of the outer ring. Together with Hertzprung and Apollo, Korolev forms a trio of huge ringed formations on the lunar far side. The lunar crust varies in thickness, and it reaches its maximum thickness of 66 miles (107 km) in the region around the Korolev Crater.



INDESTRUCTIBLE?

This Orbiter 1 image shows that later impacts have done little to obliterate the huge Korolev Crater.

EXPLORING SPACE

NUCLEAR CRATER

It is very difficult to estimate the relationship between the size of a crater and the size of the asteroid that produced it. Usually the crater is about 20 times bigger. Only in controlled nuclear explosions can an exact relationship between energy release and crater size be established. Sedan Crater in the Nevada Desert (below) is bowl-shaped and 1,200 ft (368 m) across. It was produced by a subsurface nuclear blast equivalent to 100 kilotons of TNT in July 1962. It is very similar to small lunar impact craters such as those within Korolev.

